Rohit Kumar Yadav

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Entry-level GenAI Engineer specializing in agentic AI systems, RAG pipelines, and LLM tool orchestration using LangGraph, MCP, and modern LLMOps practices. Experienced in building and deploying end-to-end AI products from prototype to production.

EDUCATION

Graphic Era University Dehradun, India | 2023 – 2025
Master of Computer Applications | CGPA: 7.4

Graphic Era University Dehradun, India | 2020 – 2023
Bachelor of Science in Information Technology | CGPA: 7.6

SKILLS SUMMARY

- **Programming & Databases:** Python, SQL, PostgreSQL, MongoDB, SQLAlchemy (Async)
- **GenAI & LLMs:** LangGraph, LangChain, AWS Bedrock, Advanced RAG, Tool Calling, MCP, LLM Agents
- **LLMOps & Evaluation:** LangSmith, DeepEval, Prompt Evaluation, RAG Evaluation
- **ML / DL:** Scikit-Learn, TensorFlow, PyTorch, Transformers, Hugging Face, LLaMA, LLM Fine-Tuning
- **NLP & Retrieval:** NLP Pipelines, Embeddings, Semantic Search, Pinecone, ChromaDB, FAISS, Vector Databases
- **Data & Visualization**: Pandas, NumPy, Matplotlib, Power BI
- **MLOps & Infrastructure:** Docker, Kubernetes, MLflow, AWS, Redis, GitHub Actions (CI/CD), Prometheus, Grafana
- **Tools & Platforms:** Git, GitHub, Jupyter Notebook, VS Code, PyCharm
- **Backend & APIs:** FastAPI, Pydantic, REST APIs, JWT Authentication, Nginx

PROJECTS

Agentic AI Assistant with MCP Integration | GitHub | Demo
Tech Stack: Python, LangGraph, LangSmith, FastAPI, MCP, Groq, Ollama, PostgreSQL, Docker, APIs

- Designed and deployed a multi-agent assistant using LangGraph orchestration, enabling autonomous execution across 7 tools (expense tracking, web search, stock analysis, weather, local events) via natural-language commands.
- Built 2 custom MCP servers (SQLite-backed) plus remote MCP integrations, extending the agent's tool-calling scope beyond a single API without hardcoding logic per tool.
- Instrumented the agent with LangSmith tracing across all tool calls, cutting average debugging time per incident from ~25 min to ~10 min by surfacing failure points in the reasoning chain instead of raw logs.

Aircraft Maintenance Intelligence System | Live | GitHub | Demo
Tech Stack: Python, FastAPI , React.js, Vite, LangChain, LangGraph, Groq, LangSmith, GitHub Actions, Vercel, Render, Docker

- Built and deployed an AI-powered aircraft maintenance platform analyzing 14 telemetry parameters, using manual-defined range thresholds for deterministic anomaly detection with Z-score and trend analysis as supporting context.
- Engineered an evidence-grounded RAG/LLM pipeline (LangGraph + Groq) retrieving maintenance-manual context to generate failure-mode reasoning and priority recommendations, scoping LLM calls to flagged anomalies to control cost.
- Productionized with FastAPI, React/Vite, and automated deployment (GitHub Actions) to Vercel/Render, with end-to-end LLM workflow tracing via LangSmith.

Multimodal AI System | Live | GitHub | Demo
Tech Stack: Python, LangChain, Streamlit, Chroma DB, Google Gemini (LLM), LangGraph, Embeddings

- Designed and deployed a multimodal RAG platform enabling natural-language querying across YouTube videos, PDFs, and documents, indexing 1,000+ chunks for semantic retrieval.
- Built a Corrective RAG (CRAG) pipeline that validates retrieved chunks before generation, reducing irrelevant/hallucinated answers by ~30% compared to a baseline retrieval-only setup.
- Built a LangGraph query router that dynamically selects the correct knowledge source (video transcript vs. PDF vs. doc) per query, reducing off-source answers by ~25%.

EXPERIENCE

Data Analytics Intern | **Infyntrek Systèmes** | **Certificate** Sep 2025 – Dec 2025

- Performed data preprocessing, feature engineering, and EDA using Python and SQL for scalable ML pipelines.
- Built, trained, and evaluated Machine Learning and Deep Learning models with Scikit-Learn and TensorFlow, applying model evaluation techniques to achieve accurate and reliable predictive performance.

HACKATHONS

- Participated in HackerRank Orchestrate, June 2026, building and presenting an AI agent in a 24-hour hackathon.
- Built RecruitRankAI for the INDIA.RUNS AI Challenge by Redrob, using Sentence Transformers, FAISS, and hybrid ranking to generate explainable Top-100 candidate recommendations under CPU-only, offline, no-LLM constraints.

CERTIFICATES

- MCP Advanced Topics (Anthropic) | Certificate
- FastAPI for Machine Learning (Campus X) | Certificate
- AI Primer Certification (Infosys) | Certificate
- Principles of Generative AI Certification (Infosys) | Certificate